# Mounir Messaoudi

PhD candidate in deep learning for biomedical image analysis | **Available from September 2027**

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## Education

**PhD in Deep Learning and Biomedical Image Analysis** 2024 – 2027 (expected)
*Inria, University of Rennes – SAIRPICO team* Rennes, France

**Thesis topic:** deep learning methods for the identification and localization of macromolecules in 3D cryo-electron tomograms.

**Publications:**

[1] **Anisotropy-Aware Strategy for Robust Deep Learning-Based Particle Picking in 3D Cellular Cryo-Electron Tomogram**
**M. Messaoudi**, C. Kervrann
*2026 IEEE 23rd International Symposium on Biomedical Imaging (ISBI)*, pp. 1–5, 2026 **Oral session**
Links: [PDF] [DOI]
Also presented at IABM 2026, Life Science PhD Symposium (3rd Edition, 2026), and AI WILD West 2026.

[2] **Designing Affine-Invariant Neural Networks for Photometric Corruption Robustness and Generalization**
**M. Messaoudi**, Q. Rapilly, S. Herbreteau, A. Badoual, C. Kervrann
*International Conference on Learning Representations (ICLR)*, pp. 124211–124271, 2026 **Poster session**
Links: [PDF]
Also presented at CAP & RFIAP 2026 and Life Science PhD Symposium (3rd Edition, 2026).

**M.Sc. in Artificial Intelligence, Systems, Data (M2 IASD)** 2022 – 2023
*Paris Dauphine University – PSL, with École Normale Supérieure and MINES Paris* Paris, France

**M.Sc. 1 in Computer Science and Engineering (Erasmus)** 2021
*Polytechnic University of Milan* Milan, Italy

**Engineering Degree, Applied Mathematics and Computer Science (Data Science)** 2017 – 2022
*CY Tech* Cergy, France

## Research and Professional Experience

**Research Intern in Computer Vision** 03/2023 – 09/2023
*Institut Curie and Centre for Computational Biology (CBIO), MINES Paris* Paris, France

- Master's thesis: deep learning method to quantify breast cancer aggressiveness from IHC images.
- Supervisor: Thomas Walter.

**Data Scientist** 09/2021 – 09/2022
*IBM Interactive, an IBM Company* Paris, France

- Graduation thesis (engineering degree): diagnostics and automatic enhancement of IBM Watson chatbots.

## Teaching

**Lectures:** *Introduction to Computer Vision* (20h per year in each programme), M2 Mathematical Engineering, CY Tech engineering school, and M2 Applied Mathematics and Statistics, University of Rennes, 2025 and 2026.
**Practical sessions:** Python, L3 Computer Science, ISTIC, University of Rennes (2025).

## Skills, Languages and Interests

- **Technical skills:** Python, PyTorch, TensorFlow, Git/GitHub.
- **Languages:** French (fluent), English (fluent).
- **Interests:** Rowing (Paris Crossing 2026; Amsterdam Tulip Crossing 2027, planned) and swimming.

## References

**Charles Kervrann** charles.kervrann@inria.fr *PhD supervisor*
**Anaïs Badoual** anais.badoual@inria.fr *Permanent member of the SAIRPICO team, co-author*
**Sébastien Herbreteau** sebastien.herbreteau@ensai.fr *Co-author*